Instructions on lengths of Final Year Project reports

All reports should be written with at least 2.0cm (0.8") margins at top, bottom and right hand side, the left hand side should be 2.5cm (1") to allow for the binding. All reports should be written in either 11 or 12 point size, 1.5 lines spacing and one of these type faces: Ariel, Times, Times New Roman or Calibri, with Symbol or similar used for Greek letters and appropriate mathematical symbols. Ariel Narrow or any other "narrow" typeface should not be used. The report can be prepared in Word, Latex or any other suitable word processing package. Pages should be numbered.

The report should contain:

Cover page

Abstract

Acknowledgements (remember to thank those people who helped you including PhD students from the group you were working with and any technicians who helped you. Also anyone who was responsible for funding the work)

List of Contents

List of Figures

List of Tables

Introduction/Literature Survey

Materials and Methods (or a similar section describing the methods used in your study)

Results

Discussion

Conclusions

(other options include Results and Discussion (one section) followed by Conclusions or Results followed by Discussion and Conclusions (one section). Also note that the Conclusions can be given as bullet points)

List of References – any sensible referencing system can be used, examples are given later in this document. Use numbers in the text to indicate where the reference is being used. The aim is that future readers can reasonably easily find the references you have used.

Appendices – extra data or any computer programs can be listed here.

Subsections within each of these sections/chapters are usually sensible.

Report lengths are measured from <u>end of contents list</u> to *beginning of references*.

MEng

The MEng report should be no more than 9,000 words and 30 figures.

BEng

The BEng report should be no more than 40 pages and 20 figures. Also, it must be no less than 30 pages.

<u>**References**</u> should be mainly be taken from journal papers relevant to the area of your project. Your supervisor will probably have given you one or two to start from, it is then part of your project work to find further references working from these. The internet is not an acceptable source of information for a project. You may also have started with a chapter from a book and only a limited number of book chapters are acceptable. Referencing from a journal should give the name of the journal volume (possibly the number within the volume) and pages NOT the internet access. If for example you search for the paper title "Effects of test sample shape and surface production method on the fatigue behaviour of PMMA bone cement", you will find one internet option of

http://www.ncbi.nlm.nih.gov/pubmed/24070780

this is not to correct the reference for this paper. It should be listed in the list of references as

E.M. Sheafi and K.E. Tanner (2014) "Effects of Test Sample Shape and Surface Production Method on the Fatigue Behaviour of PMMA Bone Cement", *Journal of Mechanical Behavior of Biomedical Materials*, Vol. 29(1), pp. 91-102.
Although the journal title and numbering could be abbreviated to *J Mech Behav Biomed Mat*, 29: 91-102. or similar. The internet link may vanish with time, the journal name with volume and pages will last as long as libraries last.
Less than 20 references (approximately) will probably indicate that you have not seriously read around the subject, more than 80 references (approximately) and it seems unlikely that you have read them all. You should have read **ALL** your references.
You probably will find using a referencing management system helpful, e.g. Reference Manager, and this should put your references in a sensible order and logical method. However, you do need to feed the information into such a system corrected otherwise you will get rubbish out when you print out the report.

All Figures should be captioned below the Figure. The caption should explain what is seen in the figure and this caption should be "stand alone" from the surrounding text. Each figure should be after the paragraph where it is first referenced it must have a reference in the text. If the figure has any taken from other people's work such as in the Literature Survey in **MUST** have the source given. It preferably should be from journal papers or books rather the internet, however if taken from the internet the date you accessed the internet should be given. If you take information/figures/tables from other sources and do not reference this is plagiarism.

All Tables should be captioned above the Table. The caption should explain what is in the Table and again caption should be "stand alone" from the surrounding text. Each table should be after the paragraph where it is first referenced and it must have a reference in the text. If the table has any taken from other people's work such as in the Literature Survey in **MUST** have the source given. It preferably should be from journal papers or books rather the internet, however if taken from the internet the date you accessed the internet should be given. If you take information/figures/tables from other sources and do not reference this is plagiarism.

Examples of how to reference figures/tables.
"Bloggs et al (2014) plotted stress strain graphs for the five composites they developed (Figure 2) and found that increasing the amount of filler …". "The numbers of lines of C++ written could be reduced by …. (Table 4)" or "Table 4 shows that numbers of lines of C++ written could be reduced by …. "The nano structures produced were viewed in the SEM (Figure 7) and showed….".
with the captions such as:
"Figure 2 Stress strain graphs for PLA and when reinforced with 5, 10, 15 or 20% of modified hydroxyapatite, showing the increase in modulus and reduction in strain to failure with increasing hydroxyapatite content (taken from Bloggs et al., 2014)."
"Table 4 Number of lines of C++ coding needed to solve Eqn 4.3 before and after the application of Newton's equation to modify the analysis procedure. "

Equations should be numbered at the end of the line containing the equation. There is no need to list the equations in the contents list.

If placing extra figures in appendices, do not use this to just add extra figures the appended figures should have a very good rationale for being in the appendix. Your reader/examiners do not guarantee to read appendices. The figures you do include in the body of the text should be clear, makes sure that they are well captioned and the colour/shading of any lines or histogram boxes are clear to the reader without needing to use a magnifying glass.

Information needed for References / Bibliography

for a journal paper:
author (s), 'title of the paper', *name of the journal*, volume number, starting page and last page, and year.
e.g. Stephenson, S 'Fuel Cells' *Motor Age* 56, 46-49, 2000

for a book:
author(s), *title*, publisher, place and year of publication.
e.g. Riley, R. Q. *Electric and Hybrid Vehicles: An Overview of The Benefits, Challenges, and Technologies,* Michigan State University, Michigan, 1999.

for a website:
author(s), 'full title of document', [the full URL], date of publication, OR last revision OR when consulted.
e.g. US Government, 'Hybrid Electric Vehicle Program', www.ott.doe.gov/hev/2000 accessed 20/1/2015

NB Take the page number too if you want to use exact information from a table or graph, or exact words as a quotation.

Words to use for putting sources into your text.
In these examples superscript numbers in the text refer to the numbers in the bibliography. You could use bracketed numbers instead, e.g. [1]

a) **One application of AI is where** robot control algorithms have been developed to locate the centre of a chemical plume by following the plume upwind[1].

b) **One advantage** of using a system which combines biological and mechanical parts is that it can conduct a real world experiment rather than a computer simulation[1].

c) **Kuwana and Shimoyama claim that** their robot might be able to detect gas leaks caused by earthquake damage[1].

d) **According to** Kuwana and Shimoyama[1] the simple control algorithm and structure of the robot minimises space requirements.

e) **Filipetti et al[2] report on** A I techniques that can be applied to fault diagnosis of induction motor drives. These may involve expert systems, artificial neural networks and fuzzy logic systems that can be integrated into each other and more traditional systems.

f) Recent developments in the field of AI based diagnostic systems in electrical machines and drives include expert systems, fuzzy logic, ANNs and combined systems, e.g. fuzzy- NNs[2].

This report shows that, although there are few practical implementations at present, these techniques will play a more significant role in the future, e.g. fuzzy neural diagnostic systems, and GA assisted neural and fuzzy neural systems for machine condition monitoring.[2]

References

1) Kuwana, Y & Shimoyama, I, 'A pheromone-guided mobile robot that behaves like a silkworm moth with living antennae as pheromone sensors'. *International Journal of Robotics Research*, 17 (9):924-933 (1998).

2) Filipetti, F, Franceschini, G, Tassoni, C & Vas, P, 'Recent developments of induction motor drives fault diagnosis using AI techniques'. *IEEE Transactions on Industrial Electronics*, 47(5): 994-1004 (2000).

Also

- Use numbers in the order in which citations appear in text
- Do not put the same source twice in reference list. If you refer to the same source twice, use the same number in the text.